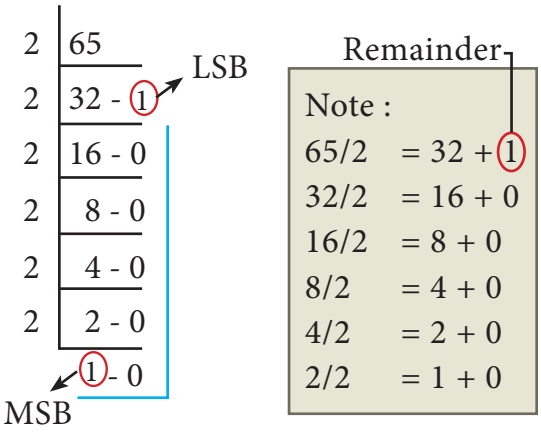


$(65)_{10} = (1\ 0\ 0\ 0\ 0\ 0\ 1)_2$

Method 2 : Sum of Powers of 2.

A decimal number can be converted into a binary number by adding up the powers of 2 and then adding bits as needed to obtain the total value of the number.

a) Find the largest power of 2 that is smaller than or equal to 65.

$65_{10} > 64_{10}$

b) Set the 64's bit to 1 and subtract 64 from the original number

65-64=1

c) 32 is greater than the remaining total. Therefore, set the 32's bit to 0.

d) 16 is greater than the remaining total. Therefore, set the 16's bit to 0.

e) 8 is greater than the remaining total. Therefore, set the 8's bit to 0.

f) 4 is greater than the remaining total. Therefore, set the 4's bit to 0.

g) 2 is greater than the remaining total. Therefore, set the 2's bit to 0.

h) As the remaining value is equivalent to 1's bit, set it to 1.

1-1=0

Conversion is complete $65_{10} = (1000001)_2$

Example

The conversion steps can be given as follows:

Given Number : 65

Equivalent or value less than power of 2 is : 64

(1) 65 - 64 = 1

(2) 1 - 1= 0

Power's of 2	64	32	16	8	4	2	1
Binary Number	1	0	0	0	0	0	1

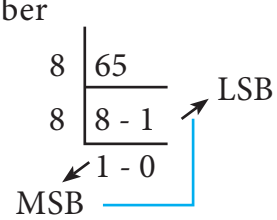
$65_{10} = (1000001)_2$

2.4.2 Decimal to Octal Conversion

To convert Decimal to Octal, "**Repeated Division by 8**" method can be used. The method is the same we have learnt in 2.4.1, but in this method, we have to divide the given number by 8.

Example

Convert $(65)_{10}$ into its equivalent Octal number



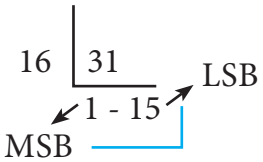
$(65)_{10} = (1\ 0\ 1)_8$

2.4.3 Decimal to Hexadecimal Conversion

To convert Decimal to Hexadecimal, "**Repeated division by 16**" method can be used. The method is the same as we have learnt in 2.4.1, but in this method, we have to divide the given number by 16.

Example

Convert $(31)_{10}$ into its equivalent hexadecimal number.



$(16)_{10} = (1F)_{16}$(Refer Table 2.2 F=15)